

Day 1 - AWS Cloud Setup and Access Management

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EC2 Instance Creation and Configuration :

I started the assignment by logging into the AWS Management Console with my AWS account. From the dashboard, I opened the EC2 service, which is used to create and manage virtual servers in the cloud.

I then selected the option to launch a new instance and named it Assignment-EC2. For the operating system, I chose the Ubuntu AMI because it is a lightweight and commonly used Linux distribution. To keep the setup cost-effective, I selected the t3.micro instance type, which is eligible for the AWS Free Tier.

For secure access to the instance, I created a key pair named Assignment-KeyPair. AWS generated a private key in .pem format, which I downloaded and stored safely. This key is needed when connecting to the EC2 instance through SSH.

Next, I configured a security group for the instance. I allowed SSH traffic on port 22 only from my IP address and enabled HTTP traffic on port 80 so that web access could be used if needed later. I kept the default 8 GB General Purpose SSD (gp3) storage because it was enough for the assignment.

After checking the configuration, I launched the instance and waited for its status to change to Running. I also verified that the system and instance status checks had passed. This confirmed that the EC2 instance was successfully deployed and working as expected.

Setting	Configuration
Instance Name	Assignment-EC2
Operating System	Ubuntu
Instance Type	t3.micro
Storage	8 GB gp3
Security Group	SSH (Port 22) from my IP address
HTTP Access	Port 80 from Anywhere

Creating an S3 Bucket with a Lifecycle Policy :

Once the EC2 setup was complete, I moved on to Amazon S3, which is AWS's object storage service. I opened S3 from the AWS Console and selected the option to create a new bucket.

I entered a globally unique bucket name and selected the required AWS region. Since the assignment did not require any special bucket settings, I kept the remaining options at their defaults and created the bucket.

To make sure the bucket was working correctly, I uploaded a sample text file and confirmed that the upload was successful. I then moved to the bucket's Management section to set up a lifecycle policy.

I created a lifecycle rule named DeleteOldFilesRule and applied it to all objects in the bucket. The rule was configured to automatically expire and delete objects after 30 days.

This was useful to understand because lifecycle policies can reduce storage costs by automatically clearing files that are no longer needed. In a production environment, this can be especially helpful when a large amount of data is stored over time.

Setting	Configuration
Bucket Name	assignment-storage-bucket-12345
Lifecycle Rule	DeleteOldFilesRule
Lifecycle Action	Automatically delete objects
Expiration Period	30 days

Creating a Least-Privilege IAM Policy :

The final task for Day 1 was to work with AWS Identity and Access Management (IAM) and apply the principle of least privilege. The idea was to give access only to the resources and actions that were actually needed.

I opened the IAM service and created a custom policy using the JSON editor. The policy allowed users or roles to list the contents of the S3 bucket, upload objects, and read existing objects. I intentionally did not include permission to delete objects or access unrelated AWS resources.

Using this approach helped me understand how limiting permissions can reduce security risks and prevent accidental changes. The custom policy was then attached to a new IAM role named Assignment-S3-Role.

Finally, I associated the role with the EC2 instance. This allows the instance to access the S3 bucket securely without having to store AWS credentials directly on the server.

Day 1 Takeaway :

By the end of Day 1, I had worked through the basic setup of an EC2 instance, created an S3 bucket with an automatic cleanup rule, and configured a least-privilege IAM role for secure access. The practical setup helped me understand how these AWS services can work together rather than learning them only as separate concepts.